

# softstart\_\_routed

## Design Report

kicad-tools 0.13.0

Rev 1 | 2026-06-10 | jlcpb

### Board Summary

| Property   | Value                                 |
|------------|---------------------------------------|
| Layers     | 4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu) |
| Footprints | 78 (0 SMD, 0 THT, 78 other)           |
| Nets       | 41                                    |
| Traces     | 1004 segments                         |
| Vias       | 104                                   |
| Board Size | 150.0 x 100.0 mm                      |

### Design Overview

#### Theory of Operation

Generator Soft-Start - Supercapacitor Power Assist

120VAC soft-start for 8000 BTU AC on Honda EU1000i

STM32G031K8T6 MCU, 2x30S supercap banks, back-to-back FETs + UCC27211 drivers

#### Power Architecture

**Power Rails:** +3V3, GND, PWR\_FLAG

| Regulator | Device      |
|-----------|-------------|
| U4        | XC6206-3.3V |
| U9        | LM7812      |

### Assembly Notes

1 fine-pitch component; 6 polarized components

- **Fine-pitch components:** 1 (U1)
- **Polarized components:** 6 – check orientation markings

## ERC Status

| Metric   | Count |
|----------|-------|
| Errors   | 0     |
| Warnings | 0     |

**Status:** SKIPPED – ERC skipped by user request

## Bill of Materials

| Value     | Package                                           | Qty | References                          |
|-----------|---------------------------------------------------|-----|-------------------------------------|
| 100nF     |                                                   | 5   | C5, C20, C22, C23, C25              |
| 100nF     | C_0402_1005Metric                                 | 8   | C1, C2, C3, C8, C30, C31, C33, C34  |
| 10uF      |                                                   | 2   | C21, C24                            |
| 10uF      | C_0805_2012Metric                                 | 3   | C6, C7, C32                         |
| 4.7uF     | C_0805_2012Metric                                 | 1   | C4                                  |
| RB157     | Diode_Bridge_DIP-4_W7.62mm_P5.08mm                | 1   | D1                                  |
| SMBJ18A   | D_SMB                                             | 4   | D_TVS1, D_TVS2, D_TVS3, D_TVS4      |
| STATUS    |                                                   | 1   | D2                                  |
| 15A       |                                                   | 1   | F1                                  |
| AC_INPUT  | TerminalBlock_bornier-2_P5.08mm                   | 1   | J1                                  |
| AC_OUTPUT | TerminalBlock_bornier-2_P5.08mm                   | 1   | J2                                  |
| SCAP_NEG  | TerminalBlock_bornier-2_P5.08mm                   | 1   | J4                                  |
| SCAP_POS  | TerminalBlock_bornier-2_P5.08mm                   | 1   | J3                                  |
| SWD-6     |                                                   | 1   | J5                                  |
| 2N7002    | SOT-23                                            | 2   | Q7, Q8                              |
| AO3400    | SOT-23                                            | 2   | Q5, Q6                              |
| IRFB4110  | TO-220-3_Vertical                                 | 4   | Q1A, Q1B, Q2A, Q2B                  |
| 100R      | R_Axial_DIN0617_L17.0mm_D6.0mm_P2.54mm_Horizontal | 2   | R2, R10, R11, R23, R24, R26, R28    |
| 10k       |                                                   | 7   | R2, R10, R11, R23, R24, R26, R28    |
| 10k       | R_0805_2012Metric                                 | 6   | R_GB1, R_GB2, R_GB3, R_GB4, R5, R29 |
| 150R 5W   | R_Axial_DIN0617_L17.0mm_D6.0mm_P2.54mm_Horizontal | 2   | R12, R22                            |
| 1k        |                                                   | 2   | R12, R22                            |
| 275VAC    | RV_Disc_D12mm_W4.2mm_P7.5mm                       | 1   | RV1                                 |
| 290.0k    |                                                   | 2   | R25, R27                            |
| 33k       | R_0805_2012Metric                                 | 2   | R3, R4                              |
| 5mR       | R_2512_6332Metric                                 | 1   | R9                                  |
| 990.0k    |                                                   | 1   | R1                                  |
| RESET     |                                                   | 1   | SW1                                 |
| H11AA1    | DIP-6_W7.62mm                                     | 1   | U2                                  |
| INA180A3  | SOT-23-5                                          | 1   | U3                                  |
| LM393     | SOIC-8_3.9x4.9mm_P1.27mm                          | 3   | U7, U7, U7                          |
| LM7812    | TO-220-3_Vertical                                 | 1   | U9                                  |
| MCP6001   | SOT-23-5                                          | 1   | U8                                  |

| Value         | Package                      | Qty | References |
|---------------|------------------------------|-----|------------|
| STM32G031K8T6 | LQFP-<br>32_7x7mm_P0.8mm     | 1   | U1         |
| UCC27211      | SOIC-<br>8_3.9x4.9mm_P1.27mm | 2   | U5, U6     |
| XC6206-3.3V   |                              | 1   | U4         |

## DRC Status

| Metric   | Count |
|----------|-------|
| Errors   | 7     |
| Warnings | 10    |
| Blocking | 7     |

**Status:** FAIL ### Violations by Type

| Violation Type        | Count |
|-----------------------|-------|
| pad_grid              | 8     |
| via_in_pad            | 6     |
| connectivity          | 2     |
| zone_unfilled         | 2     |
| clearance_pad_segment | 1     |

## Manufacturing Readiness

Verdict: NOT\_READY

### Action Items

- **[CRITICAL]** Fix 7 blocking DRC violations (via\_in\_pad (6), clearance\_pad\_segment (1))
- **[OPTIONAL]** Verify zone fill in KiCad: 2 nets appear incomplete but may be connected via zone fills
- **[OPTIONAL]** Verify zone fill in KiCad for 13 zone-connected nets
- **[OPTIONAL]** Review 10 DRC warnings
- **[OPTIONAL]** Analog-sensitive: U3 (INA180A3) — instrumentation amplifier (TI INA); manual layout review recommended
- **[OPTIONAL]** Analog net: AC\_LINE — audio signal; keep short, away from digital/switching nets
- **[OPTIONAL]** Analog net: BUS\_LINE — audio signal; keep short, away from digital/switching nets
- **[OPTIONAL]** Analog net: FUSED\_LINE — audio signal; keep short, away from digital/switching nets
- **[OPTIONAL]** Analog net: ISENSE\_NEG — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: ISENSE\_POS — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: I\_SENSE\_OUT — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: V\_AC\_SENSE — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: V\_AC\_SENSE\_RAW — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: V\_BANK\_NEG\_SENSE — analog signal; noise-sensitive, avoid crossing digital signals
- **[OPTIONAL]** Analog net: V\_BANK\_POS\_SENSE — analog signal; noise-sensitive, avoid crossing digital signals

### Analog Components

1 analog-sensitive component detected – manual layout review recommended.

| Reference | Value    | Reason                             |
|-----------|----------|------------------------------------|
| U3        | INA180A3 | instrumentation amplifier (TI INA) |

## Routing Status

| Metric                | Value                 |
|-----------------------|-----------------------|
| Signal Net Completion | 92.3% (24/26)         |
| Overall Completion    | 95.1%                 |
| Complete Nets         | 39 / 41               |
| Zone-Connected Nets   | 15                    |
| Single-Pad Nets       | 2 (no routing needed) |
| Incomplete Nets       | 2                     |
| Unconnected Pads      | 8                     |

### Zone-Connected Nets

- +3.3V
- AC\_LINE
- AC\_NEUTRAL
- BUS\_LINE
- FUSED\_LINE
- GND
- ISENSE\_POS
- SCAP\_NEG+
- SCAP\_NEG\_GND
- SCAP\_POS+
- SCAP\_POS\_GND
- SRC\_NEG
- SRC\_POS
- VGATE
- VRECT

### Single-Pad Nets

2 single-pad nets (no routing needed) – not listed individually.

### Unrouted Signal Nets

- NRST
- V\_BANK\_POS\_SENSE

### Unrouted Signal Nets

- NRST
- V\_BANK\_POS\_SENSE

## Cost Estimate

| Metric          | Per Board (estimated) |
|-----------------|-----------------------|
| PCB Fabrication | ~5.0 USD              |

| Metric                   | Per Board (estimated) |
|--------------------------|-----------------------|
| Components (estimated)   | ~5.74 USD             |
| Assembly (estimated)     | ~2.52 USD             |
| <b>Total (estimated)</b> | <b>~13.26 USD</b>     |
| Batch Quantity           | 5                     |
| Batch Total (estimated)  | ~66.29 USD            |